

Multicopter Power Simulator — Performance Analysis

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Component Weight Summary

Component	Unit (g)	Qty	Total (g)	%
Battery	4200.0	1	4200.0	31.1
Motor	410.0	8	3280.0	24.3
ESC	95.0	8	760.0	5.6
Propeller	110.0	8	880.0	6.5
Payload	3000.0	1	3000.0	22.2
Airframe / Structure	1380.0	1	1380.0	10.2
TOTAL			13500.0	100.0

Design Inputs

Parameter	Value	Parameter	Value
Num Motors	8	Motor Pole Count	14
Weight	10500	Motor Weight	410
Payload Mass G	3000	Motor Size	8318
Speed	8	Esc Voltage Rating	12
Periph Current	0	Esc Cont Current	60
Body Length M	0.45	Esc Max Current	80
Body Width M	0.4	Esc Idle Current	0.05
Body Height M	0.2	Esc R	0.002
Arm Length M	0.4	Esc Weight	95
Arm Width M	0.03	Avionics Voltage Tree	5:(2,0.9), 12:(2.5,0.87), 24:(1,0.9)
Coaxial Spacing M	0.1	Prop D	22
Max Tilt Deg	20	Prop Pitch	7.2
Motor Configuration	coaxial	Prop Blades	2
Drag Model Mode	auto	Prop Max Rpm	0
Batt Vmin	3.3	Prop Max Thrust	9500
Batt Vnom	3.7	Prop Table	C:/Users/jclom/OneDrive/Desktop/
Batt Vmax	4.2	Prop Weight	110
Batt Unit Mode	pack	Mission	C:/Users/jclom/OneDrive/Desktop/
Batt Pack Capacity	16000	Alt	0
Batt C Cont	15	Temp	20
Batt Dischg Pct	80	Wind	0
Batt R	3.0	Wind Dir	0
Batt Chem	LiPo	Course Deg	0
Batt Soc Model	auto	Climb Rate	0
Batt Series	2	Descent Rate	0
Batt Parallel	1	Reserve Percent	20
Batt Cells Series	6	Rth Reserve Wh	0
Batt Cells Parallel	1	Diversion Reserve Wh	0
Batt Pack Weight	2100	Transient Dt S	0.5
Motor Kv	170	Max Accel Mps2	2.0
Motor I0	0.6	Max Decel Mps2	2.5
Motor V0	10	Decel Regen Eff	0
Motor R	0.045	Inflow Map Enabled	0
Motor Imax	45	Orientation	forward
Motor Pmax	2000	Max Speed Plot	25

Performance Metrics

Metric	Value	Metric	Value
■■ Battery ■■		■■ Flight Performance ■■	
■■ Motor @ Operating Point ■■		■■ Thermal Estimates ■■	
■■ Total Drive & Power ■■		■■ Environment & Design ■■	
■■ Propeller & Rotor ■■			

Status Checks

Battery Status

Metric	Value	Limit	Notes
Pack voltage (loaded)	46.68 V	>= 39.60 V	Sag: 3.72 V (7.4%)
Tilt req vs max	0.0° / 20.0°	<= 20.0°	
Pack current vs cont rating	33.87 A	<= 240.00 A	
Pack current vs max rating	33.87 A	<= 240.00 A	
Discharge C-rate	2.12 C	<= 15.0 C (cont)	Continuous C-rate at this speed
Total electrical power	1633 W	—	
Power split (Motor/ESC/Av)	94% / 2% / 4%	—	Motor propulsion / ESC losses / Avionics
Energy reserve margin	+380.1 Wh	>= 0 Wh	OK

Motor / ESC Status

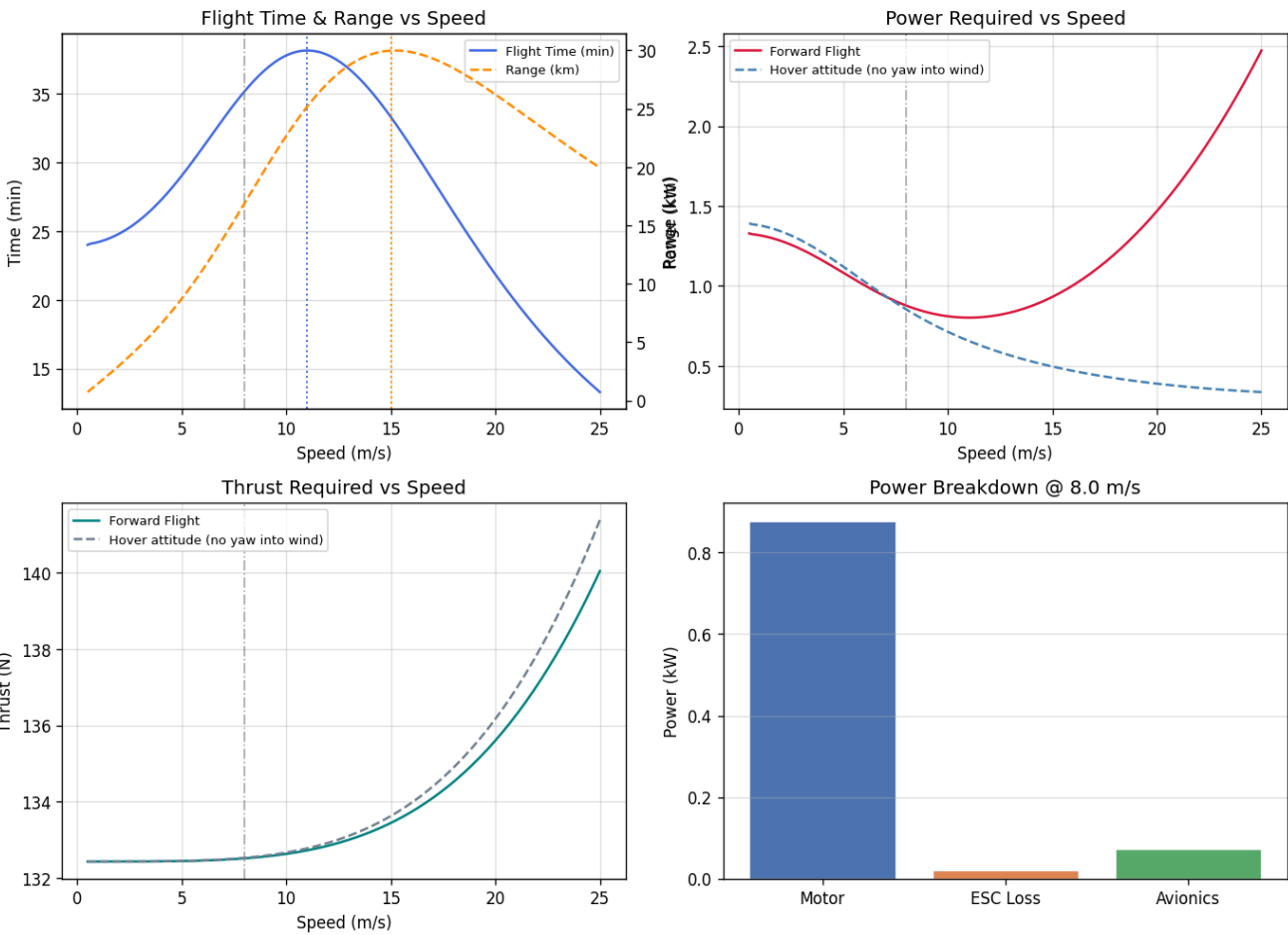
Metric	Value	Limit	Notes
Motor power / motor	191 W	<= 2000 W	
Motor current / motor (est)	3.94 A	<= 45.00 A	
Thrust-to-weight (max available)	3.81:1	>= 2.0:1	< 1.5 is dangerously marginal; >= 2.0 gives control authority
Max additional payload	37870 g	>= 0 g	Extra mass liftable at TWR 1.0, i.e. with no hover margin
Payload at TWR 2.0	12185 g	>= 0 g	Extra mass while keeping a 2:1 margin — the usable figure
Thrust required / weight	1.00:1	~1.0:1	Sits at 1 in steady flight by definition; rises with tilt in forward flight
Motor temperature (est)	21.0 °C	<= 100 °C	Headroom: 99.7 °C OK
ESC temperature (est)	31.8 °C	<= 90 °C	
Battery temperature (est)	24.5 °C	<= 55 °C	
ESC voltage rating	12S (50.4 V max)	>= 12S (50.4 V)	ESC S-rating must meet or exceed the pack series count
ESC current vs cont	3.94 A	<= 60.00 A	
ESC current vs max	3.94 A	<= 80.00 A	
ESC total loss	25.2 W	—	

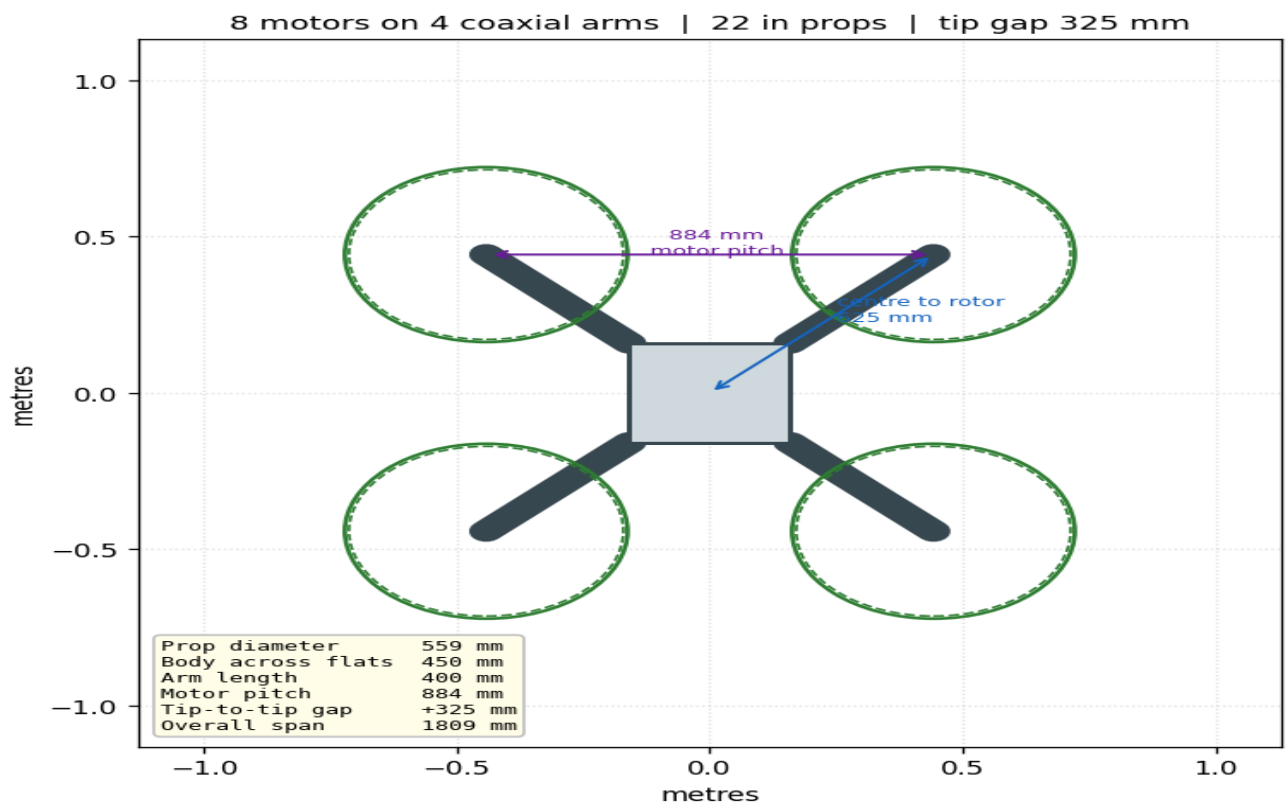
Propeller Status

Metric	Value	Limit	Notes
Thrust / motor	1691 g	<= 9500 g	Margin: 7809 g
Prop RPM (est)	2865 rpm	—	
Tip Mach	0.246	<= 0.60	Above 0.6 → significant aeroacoustic noise & efficiency loss
Disk loading	67.5 N/m²	<= 200 N/m²	High DL → less efficient hover; < 100 N/m² is good
Hover efficiency	9.57 g/W	>= 5 g/W	Higher = more lift per watt; typical 5–12 g/W
Figure of merit	0.498	>= 0.65	Ideal = 1.0; well-designed prop 0.7–0.8
Max hover wind resistance	48.8 m/s (94.8 kt)	>= 5 m/s	Maximum wind the vehicle can hold position against
Table thrust range	operating at 1691 g (100% of table value)	1685–1641 g (measured)	Far below the tested range — table values are extrapolated
Prop solidity σ	0.140	0.05 – 0.15	< 0.05 → lightly loaded blades; > 0.15 → high profile losses

Performance Plots

Multicopter Performance





Simulation Output Log

=== Mission: mc_02_takeoff_square_land.json ===

Orientation: forward | Wind: 0.00 m/s @ 0.0°

Takeoff climb: 0.42 min, 0.00 km [OK]

Leg 1 north: 0.71 min, 0.40 km [OK]

Leg 2 east: 0.67 min, 0.40 km [OK]

Leg 3 south: 0.67 min, 0.40 km [OK]

Leg 4 west: 0.67 min, 0.40 km [OK]

Hover to position: 0.33 min, 0.02 km [OK]

Descend and land: 0.67 min, 0.00 km [OK]

TOTAL: 4.12 min, 1.62 km

Reserve target/margin: 113.7 / +380.1 Wh

SoC (min/model): 86.9% / preset:lipo

Transient (dt/a+/a-): 0.50s / 2.00 / 2.50 m/s²

Inflow $\mu/\eta/\text{mult}$: 0.119 / 1.000 / 1.000

Thermal M/ESC/B: 21.0 / 31.8 / 24.5 °C [OK]

Sensitivity

Input	-20%	-10%	baseline	+10%	+20%	swing
All-up weight	49.30	41.36	35.15	30.15	26.33	22.97 (65%)
Battery capacity	28.12	31.64	35.15	38.67	42.18	14.06 (40%)
Prop diameter	31.75	33.41	35.15	36.92	38.67	6.92 (20%)
Cruise speed	31.92	33.60	35.15	36.46	37.44	5.52 (16%)
Parasite area	35.62	35.39	35.15	34.92	34.69	0.93 (3%)
Motor Kv	35.15	35.15	35.15	35.15	35.15	0.00 (0%)
Motor resistance	35.15	35.15	35.15	35.15	35.15	0.00 (0%)
Avionics draw	35.15	35.15	35.15	35.15	35.15	0.00 (0%)

Comparison against pinned baseline

Metric	Baseline	Current	Change	Change %
Flight time (min)	—	—	—	—
Range (km)	—	—	—	—
Total power (W)	2481.0	1632.8	-848.2	-34.2 %
Pack current (A)	52.72	33.87	-18.86	-35.8 %
Loaded voltage (V)	45.28	46.68	+1.40	+3.1 %
Total thrust (N)	181.63	132.63	-49.00	-27.0 %
Hover efficiency (g/W)	8.34	9.57	+1.22	+14.6 %
Figure of merit	0.509	0.498	-0.010	-2.1 %
Disk loading (N/m²)	92.5	67.5	-25.0	-27.0 %
Motor temp (°C)	22.7	21.0	-1.7	-7.4 %
Reserve margin (Wh)	337.47	380.09	+42.62	+12.6 %